

Hardware CWE SIG Meeting Notes

Meeting Minutes

May 15, 2026

Meeting Attendance

Sohrab Aftabjahani	Amitabh Das	Jeremy Lee
Irena Bojanova	Alexander Harner	Jason Oberg
Steve Christey	Bob Heinemann	Mitchell Poplingher

Agenda

- Housekeeping and announcements
- RTL Working Group progress update
- Review of follow-up from the prior discussion on CWE relationships
- Discussion of hardware view inconsistencies, including parent-child relationship presentation
- Discussion of category assignment issues and related quality assurance considerations

Meeting Notes

Housekeeping and Announcements

Bob Heinemann opened the meeting with a brief review of the agenda and noted that the next Hardware CWE SIG meeting is scheduled for June 19. He also reported that CWE 4.20 was released on April 30.

A hardware content correction was noted for CWE-1268. Bob explained that an error in a demonstrative example involving bit masking had been identified and corrected, and he thanked Jagadish for bringing the issue to the team's attention. Bob also reminded attendees that the content development repository is available for tracking submissions and reviewing content that is in progress.

RTL Working Group Progress

Bob and Mitchell Poplingher provided an update on the RTL Working Group. The group met on April 20 and again during the week of this meeting. They reported strong collaboration both during meetings and through email exchanges.

The working group has developed what was described as a strong draft of the extended description, following multiple iterations. The group indicated that it is satisfied with the current draft and has shifted its attention to additional required elements. Current drafting priorities are summary, common consequences, mitigations, and multiple examples. Working drafts for these sections are in progress, although they are at different levels of completion.

Follow-up on Prior Relationship Discussion

Bob briefly revisited the prior month's discussion led by Jason Oberg regarding clearer definition of relationships. He reported that the MITRE team, in consultation with Jason, created a submission in the content development repository to capture the proposed content changes for HW relationships. Participants were encouraged to review that submission and provide additional comments or suggestions there, with the intent of incorporating changes into a future CWE release.

Hardware View Inconsistencies Discussion

The primary discussion focused on inconsistencies between how relationships appear in the research view and how they are presented in the hardware view. Bob explained that the hardware view serves as an important entry point for the hardware community, and that inconsistencies in relationship presentation could create ambiguity for users.

Bob outlined four issue types that had been identified. The first two were the main discussion topics: parent-child relationships that appear in the research view but are flattened in the hardware view, and cases where related parent and child CWEs appear in

different hardware categories. He also noted two additional issues described as content encoding or rendering problems that the team expected to correct separately.

Using examples including CWE-325 and CWE-1240, as well as CWE-1302 and CWE-1294, Bob showed how parent-child relationships that exist in the research view may not be visible in the hardware view. He asked whether the hardware view should mirror those relationships or whether a flatter presentation was intentional and preferable.

Discussion reflected differing perspectives. One view favored consistency with the broader CWE hierarchy, noting that CWE is fundamentally structured around abstraction levels and that preserving those relationships helps users move between more general and more specific weaknesses. Another suggested that flattening entries may help ensure that each hardware CWE receives distinct attention, with related weaknesses potentially called out separately rather than through explicit parent-child hierarchy. Jeremy Lee noted that, regardless of the final approach, consistency in presentation would be important.

Irena Bojanova emphasized that hierarchical structure is a foundational aspect of CWE and argued that hardware-related weaknesses should remain consistent with that larger structure, especially since vulnerabilities may span hardware and software concerns. Steve Christey clarified that the research view captures actual abstraction relationships among weaknesses, whereas hardware view categories are convenience groupings rather than hierarchical weakness relationships. He also explained that CWE supports multiple parents in some cases, but that categories and parent-child abstraction serve different purposes.

Category Assignment and QA Discussion

Bob then moved to the second issue type: cases in which a child and its parent do not appear in the same hardware category. He presented this as a possible quality assurance metric rather than a strict rule. The examples discussed included CWE-1261 and CWE-1384, and CWE-1243 and CWE-1263. In some cases, attendees agreed that a parent's broader scope might make category overlap unnecessary; in other cases, missing category placement could indicate an omission that should be reviewed.

Steve noted that if a parent belongs to a category, its children may often also be relevant to that category, whereas the reverse may not always hold if the child is much more specific than the parent. Bob added that some of the observed issues may stem from relatively new hardware entries not yet being fully incorporated into appropriate categories. He indicated

that the team is exploring use of parent-child and category relationships as a QA metric for pre-release review so that missing or inconsistent placements can be identified more systematically.

The discussion also addressed terminology and presentation. Bob and Steve explained the distinction between classes in the research view and categories in the hardware view, including the iconography used to distinguish them. Categories were described as convenience-based groupings organized around hardware development concerns, while classes and related abstraction levels represent formal CWE hierarchy.

Closing and Follow-up

As the meeting approached the end of the scheduled hour, Bob said he would summarize the recommendations and viewpoints shared during the discussion and send them to the broader group by email to gather additional input, especially from regular participants who were not present. He also stated that the meeting video would be uploaded so absent members could review the discussion and provide feedback.

Decisions Made

No formal decisions were recorded.

Action Items

- Bob Heinemann: Summarize the meeting's recommendations and feedback on hardware view parent-child relationship presentation and send an email to the broader group for additional input.
- Bob Heinemann: Upload the meeting video so absent members can review the discussion and provide feedback.
- Bob Heinemann; Steve Christey: Review hardware CWE entries using parent-child and category relationships as a quality assurance metric to identify and correct category placement issues.